

Samuel Hale

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Summary

Software engineer with 3+ years building C#/.NET services, Angular front ends, and AI agent tooling on AWS and Kubernetes, with hands-on CI/CD, Terraform, and production reliability work on a mortgage-closing platform.

Skills

Languages: C#, TypeScript, JavaScript, Python, SQL, C++, Java, HTML/CSS

Frameworks: .NET 8/10, Angular, RxJS, React, Node.js/Express, NUnit, Model Context Protocol (MCP)

Databases: PostgreSQL (AWS Aurora/RDS), SQL Server, MongoDB

Cloud/Infrastructure: AWS (Aurora/RDS, ECR, Bedrock AgentCore), Kubernetes, Helm, Argo CD, Rancher, Terraform, Docker

Tools: Azure DevOps Pipelines, Git, Artifactory, Snyk, RabbitMQ, Claude Code, Jira

Experience

Software Engineer, Rocket Mortgage — Detroit, MI

Aug 2024 – Present

- Built a Claude-based code review tool that checks PR diffs against Azure DevOps acceptance criteria and coding standards, ranking findings by severity; cut review time from ~45 to 5–10 min per PR, adopted org-wide, later run in CI for 6+ teams
- Built 4 production MCP tools in C#/.NET letting an AWS Bedrock AgentCore AI agent text notaries, confirm vendors, and update closings; it scheduled 312 closings in one month with no human intervention (~33 hrs projected savings)
- Unblocked the agent's production launch by debugging failed Argo CD canary rollouts in Rancher and the Argo dashboard, then fixing flaky integration tests so CI checks could gate deploys without triggering false rollbacks
- Refactored 6 of the AI agent's MCP read tools onto a coarse-grained command pattern, replacing fragile in-process object-graph traversal to eliminate runtime serialization failures, backed by 64 new unit tests
- Upgraded 4 production Aurora PostgreSQL databases to v17 via Terraform ahead of AWS price hikes on older engines
- Migrated org-wide NPM package resolution to Artifactory in response to an active supply-chain attack campaign
- Diagnosed a 97.5% failure rate on a production integration message from 90 days of failure data and shipped the fix
- Built a PII-aware, dependency-ordered data retention and purge pipeline in PostgreSQL to meet a compliance deadline
- Contributed to lifting closing-schedule automation success from 17% to 40% (~264 more orders, ~53 staff hours/month)
- Review ~1/3 of PRs for a 5-engineer team and mentored an associate engineer toward mid-level output expectations

Software Engineer Intern, WoodWing — Detroit, MI

2023 – 2024

- Built a JavaScript/Workato automation that links summary titles to chapter headers through the Assets API, saving an estimated 100+ hours of manual editing
- Eliminated manual HTML validation with the W3C validation API and automated XML metadata archiving by article status
- Secured customer production automations with AWS Secrets Manager and ran UAT through Jira and Zendesk to deliver 40 books to the Brazilian government

Projects

StorAI | Node.js, Express, PostgreSQL, React

storai.net

- Built a Node.js/Express REST API with Passport authentication over a normalized PostgreSQL schema, plus a React dashboard tracking course progress for 20+ users
- Automated deployments with a GitHub Actions workflow that ships to a DigitalOcean Linux server, saving 20+ hours of manual deployment

PlayPal AI | Python, scikit-learn, React

github.com/ww-samuel-hale/playpal

- Built a game recommendation engine using scikit-learn cosine similarity over user and game vector profiles, served through a React front end
- Sped up recommendations by 80% with a capped FIFO structure that stores past interactions and prevents repeat suggestions

Education

B.S. Computer and Information Science | University of Michigan